

PCCST501 COMPUTER NETWORKS — HANDS-ON LAB MANUAL

Module-wise Hands-on Implementations in C & Python | Prof. Anju Markose, VJCET

MODULE 1 LAB: Wireshark HTTP / DNS & Python Socket Client

1. Wireshark Packet Sniffing: Capture HTTP GET requests and analyze headers, 200 OK responses, cookies, and cache control headers.
2. Python Web Client: Implementing basic HTTP GET / DNS resolver script using Python socket library.

```
# Python HTTP Client Example
import socket
host = "example.com"
port = 80
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((host, port))
req = f"GET / HTTP/1.1\r\nHost: {host}\r\nConnection: close\r\n\r\n"
s.sendall(req.encode())
res = s.recv(4096)
print(res.decode('utf-8', errors='ignore'))
s.close()
```

MODULE 2 LAB: TCP/UDP Sockets, I/O Multiplexing (select/poll) & Linux Routing

1. TCP Multi-Client Echo Server in C using select() / poll() system calls.
2. Elementary UDP Client-Server program.
3. Linux Routing Table Configuration: Using `ip route` commands to inspect FIB and add static routing entries.

```
// TCP Server using select() multiplexing snippet
fd_set readfds;
FD_ZERO(&readfds);
FD_SET(server_fd, &readfds);
int max_sd = server_fd;
int activity = select(max_sd + 1, &readfds, NULL, NULL, NULL);
if (FD_ISSET(server_fd, &readfds)) {
    int new_socket = accept(server_fd, (struct sockaddr *)&address, (socklen_t *)&addrlen);
    printf("New client connected on socket %d\n", new_socket);
}
```

MODULE 3 LAB: Raw Sockets (PF_PACKET), Frame Sniffing & CRC Generator

1. Linux Raw Socket Packet Sniffer using `PF\_PACKET` and `SOCK\_RAW` to capture Ethernet frames.
2. CRC Error Detection algorithm implemented in C / Python.

```
#include <stdio.h>
#include <sys/socket.h>
#include <netinet/if_ether.h>
#include <arpa/inet.h>
#include <unistd.h>
int main() {
    int sock_raw = socket(AF_PACKET, SOCK_RAW, htons(ETH_P_ALL));
    unsigned char buffer[65536];
    while(1) {
        int data_size = recvfrom(sock_raw, buffer, 65536, 0, NULL, NULL);
        struct ethhdr *eth = (struct ethhdr *)buffer;
        printf("Frame: Dest: %.2X:%.2X:%.2X:%.2X:%.2X:%.2X Proto: 0x%04X\n",
            eth->h_dest[0], eth->h_dest[1], eth->h_dest[2], eth->h_dest[3], eth->h_dest[4], eth->h_dest[5], ntohs(eth->h_proto));
        close(sock_raw);
        return 0;
    }
}
```

MODULE 4 LAB: SNMP Network Management & Line Coding Generator

1. SNMP Walk and Get in Linux using `net-snmp` utilities.
2. Line Coding Generator: Python script generating NRZ, Manchester, and AMI signal representations.

```
# SNMP Query Example using pysnmp / net-snmp command
# Shell command: # snmpget -v2c -c public 127.0.0.1 1.3.6.1.2.1.1.1.0
# snmpwalk -v2c -c public 127.0.0.1 1.3.6.1.2.1.2.1.2
```

```
(Interfaces) def line_code_manchester(bits): signal = [] for b in bits: if b == 0:  
signal.extend([1, -1]) # High to Low else: signal.extend([-1, 1]) # Low to High return signal
```